

PHYS 230 - Electricity and Magnetism

Lecture 12

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Outline

Review

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

Important Vector Operations

- Unit vector corresponding to a vector $\mathbf{F}$

$$\hat{\mathbf{F}} = \frac{\mathbf{F}}{F}$$

- Inner product of unit vectors

$$\hat{\mathbf{F}} \cdot \hat{\mathbf{F}} = 1$$

and

$$\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = 1$$

$$\hat{\mathbf{j}} \cdot \hat{\mathbf{j}} = 1$$

$$\hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = 1$$

$$\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = 0$$

$$\hat{\mathbf{j}} \cdot \hat{\mathbf{k}} = 0$$

$$\hat{\mathbf{i}} \cdot \hat{\mathbf{k}} = 0$$

- Cross products

$$\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$$

and

$$\hat{\mathbf{i}} \times \hat{\mathbf{i}} = 0,$$

$$\hat{\mathbf{j}} \times \hat{\mathbf{j}} = 0,$$

$$\hat{\mathbf{k}} \times \hat{\mathbf{k}} = 0$$

$$\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}},$$

$$\hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}},$$

$$\hat{\mathbf{k}} \times \hat{\mathbf{i}} = -\hat{\mathbf{j}}$$

Equivalent Forms of Electric Force Vector of Point Charges

Depending on the problem, one of the three equivalent form of vector electric field is used:

Form 1: If you only need the magnitude (or you can find the direction with other methods)

$$F_{12} = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r_{12}^2}$$

Form 2: Computing the unit vector $\hat{\mathbf{r}}_{12}$ and distance r separately

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{\mathbf{r}}_{12}$$

Form 3: Computing everything together

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} q_1 q_2 \frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|^3}$$

Equivalent Forms of Electric Field Vector of Point Charges

Three equivalent forms of $\mathbf{E}$ at point p (no need to have another charge at p) due to a source charge Q :

Form 1: Only magnitude of the field

$$E_{Qp} = \frac{1}{4\pi\epsilon_0} \frac{|Q|}{r_{Qp}^2}$$

Form 2: Using unit vector from the source Q towards any point p (can be a point in empty space)

$$\mathbf{E}_{Qp} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r_{Qp}^2} \hat{\mathbf{r}}_{Qp}$$

Form 3: Using full displacement vector from the source Q towards any point p (can be a point in empty space)

$$\mathbf{E}_{Qp} = \frac{1}{4\pi\epsilon_0} Q \frac{\mathbf{r}_p - \mathbf{r}_Q}{|\mathbf{r}_p - \mathbf{r}_Q|^3}$$

Charge Distributions: Densities

- Linear

$$\lambda = \frac{dq}{d\ell} \Rightarrow dq = \lambda d\ell$$

- Surface

$$\sigma = \frac{dq}{dA} \Rightarrow dq = \sigma dA$$

- Volume

$$\rho = \frac{dq}{dV} \Rightarrow dq = \rho dV$$

- If uniform distributions, then

$$\lambda = \bar{\lambda} = \frac{Q}{L} = \text{constant},$$

$$\sigma = \bar{\sigma} = \frac{Q}{A} = \text{constant}$$

$$\rho = \bar{\rho} = \frac{Q}{V} = \text{constant}$$

- For distributions, choose a suitable dq , take it as a point charge, compute its force, field, energy, potential, etc., then add the contribution of all these dq together using an integral, example

$$d\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{dq}{r_{qp}^2} \hat{\mathbf{r}}_{qp} \Rightarrow \mathbf{E} = \int \frac{1}{4\pi\epsilon_0} \frac{dq}{r_{qp}^2} \hat{\mathbf{r}}_{qp}$$

Some Special Systems

- The field above/below a sheet with surface charge densities σ , is perpendicular to the surface and

$$E = \frac{\sigma}{2\epsilon_0}$$

- The field just outside the surface of a conductor, is perpendicular to the surface and

$$E = \frac{\sigma}{\epsilon_0}$$

- Electric dipole moment

$$\mathbf{p} = q \mathbf{r}_{-+}$$

- Torque by an external $\mathbf{E}$ on a dipole with moment $\mathbf{p}$

$$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$$

- Potential energy of a dipole with moment $\mathbf{p}$ in an external $\mathbf{E}$

$$U = -\mathbf{p} \cdot \mathbf{E}$$

Flux and Gauss's Law

- Area vector (infinitesimal or finite) is perpendicular to the surface with a magnitude equal to the area

$$d\mathbf{A} = dA \mathbf{n}$$

- $d\mathbf{A}$ for closed surfaces always points outwards
- Flux $d\Phi$ of an electric field $\mathbf{E}$ through an area $d\mathbf{A}$

$$d\Phi = \mathbf{E} \cdot d\mathbf{A} = E dA \cos(\theta_{\mathbf{E}, d\mathbf{A}})$$

- Gauss's law

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{in}}}{\epsilon_0}$$

Electric Potential Energy

- If a charge q moves in an electric field, its potential energy changes as

$$dU = -dW \Rightarrow U_b - U_a = - \int_{\mathbf{r}_a}^{\mathbf{r}_b} \mathbf{F} \cdot d\mathbf{r} = -q \int_{\mathbf{r}_a}^{\mathbf{r}_b} \mathbf{E} \cdot d\mathbf{r}$$

- General expression for potential energy

$$U = -q \int \mathbf{E} \cdot d\mathbf{r}$$

- Potential energy of a system of point charges

$$U_{\text{total}} = \underbrace{\frac{1}{2} \sum_{\substack{i,j=1 \\ i \neq j}}^n U_{ij}}_{U_{12} + U_{13} + U_{23} + \dots} = \underbrace{\frac{1}{2} \left[\frac{1}{4\pi\epsilon_0} \sum_{\substack{i,j=1 \\ i \neq j}}^n \frac{q_i q_j}{r_{ij}} \right]}_{\frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} + \dots \right)}$$

Important Point!

- In any computation which involves an integral over a curve with the infinitesimal displacement $d\mathbf{r}$, regardless of the direction of the curve, DO NOT CHANGE THE DIRECTION (SIGN) OF $d\mathbf{r}$.
- **Change of integral limits** due to change of direction of movement on the curve **will take care of the sign!**